

COMPARISON BETWEEN AKI, AKD, AND CKD

00:11:52

	AKI	AKD	CKD
Duration	<7 days	<3 months	≥3 months
Functional Criteria	S. Cr ↑ by 50% in 7 days OR S. Cr ↑ by >0.3 mg/dl within 48 hrs OR - U.O <0.5 mL/kg/hr >6 hrs	- AKI OR - GFR <60 mL/min OR - GFR ↓ >35% over baseline OR ↑ in S. Cr by 50% over baseline	GFR <60 mL/min/1.73 m²BSA for >3 months
Structural Criteria	Not defined	Marker of Kidney damage → Albuminuria, Hematuria, Pyuria	Marker of Kidney damage → Albuminuria

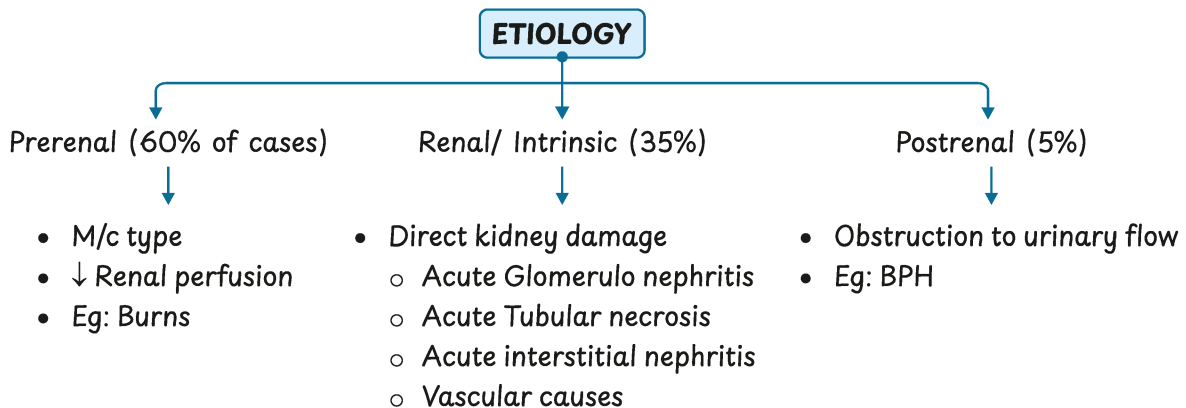
NOVEL BIOMARKERS FOR AKI

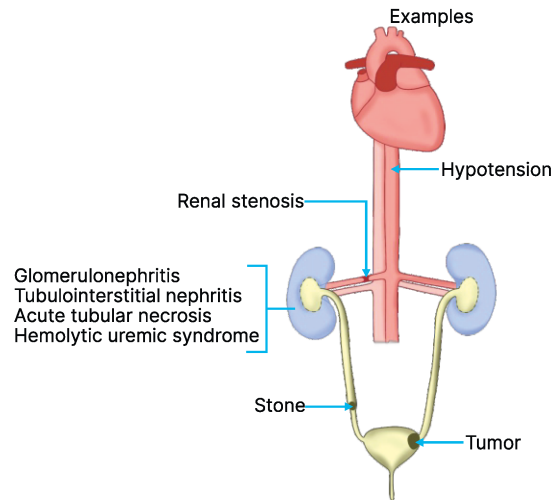
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- Urinary KIM-1 (Kidney Injury Molecule-1)
 - Values will be elevated in kidney damage
 - Rises much faster compared to S. Creatinine
- Urinary NGAL (Neutrophil Gelatinase Associated Lipocalin/ Lipocalin-2)
- Serum Cystatin-C
 - More reliable than creatinine as it is freely filtered across glomerulus and not secreted/ reabsorbed across tubules
- sUPAR (Soluble Urokinase Plasminogen Activator Receptor)
- Osteopontin
- Clusterin

ETIOLOGY

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I. PRERENAL AZOTEMIA

CAUSES

- Hypovolemia: Burns, cholera, heat stroke, bleeding – trauma / obstetric
- CHF/ ischemic Cardiomyopathy → ↓ Cardiac output → ↓ Renal perfusion
- Cirrhosis (Hepatorenal syndrome)
 - Cirrhosis causes production of Nitric oxide → ↓ SVR → ↓ Renal perfusion
- Drugs:
 - COX-1 inhibitors → ↓ PGs → ↓ Renal perfusion
 - ACE inhibitors/ARBs → Block angiotensin II → EA dilation → Loss of pressure gradient
 - Cyclosporine → Liberates Nitric oxide → ↓ Renal perfusion

Important Information

- Afferent arteriole: Vasodilation is maintained by PGs
- Efferent arteriole: Vasoconstriction is maintained by Angiotensin II

II. RENAL/INTRINSIC CAUSES

GLOMERULAR	TUBULAR/INTERSTITIUM	VASCULAR
• Acute Glomerulonephritis ◦ RBC casts ◦ Dysmorphic RBCs	• Sepsis: ↓ Cardiac output, ↓ Tubular blood supply • Ischemia ◦ Prolonged anesthesia • Nephrotoxins ◦ Contrast induced nephropathy ◦ Ethylene glycol, and Diethylene glycol → Adulterant in cough syrup	• HUS: ◦ E. Coli O157- H7 produces Shiga toxin → Toxin mediated compliment activation → Renal shutdown • TTP ◦ ↓ ADAMTS13 → Platelet rich microthrombi • Malignant HTN → End organ damage • Atheroembolization Kidney disease ◦ Urine M/E: Eosinophiluria • Renal vein thrombosis • Renal artery embolus



Important Information

→ 25 segment of PCT (present on renal outer medulla) is most vulnerable to injury d/t high O_2 consumption

ISCHEMIC INJURY TO TUBULES

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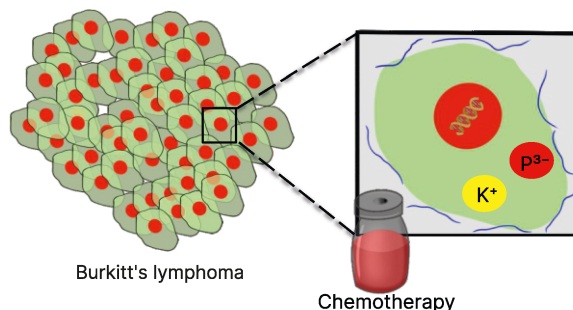
1. BURNS/ACUTE PANCREATITIS	Massive fluid resuscitation ↓ Cytokines causing leaky capillaries ↓ Abdominal compartment syndrome
2. PROLONGED INTRAOPERATIVE HYPOTENSION	• Cardiopulmonary bypass → Has a non-pulsatile flow • Aortic cross-clamping
3. THROMBOTIC MICROANGIOPATHY (TMA)	• Drugs responsible: ○ Clopidogrel ○ Mitomycin-C ○ Quinine (T/t of malaria)

NEPHROTOXINS

EXOGENOUS NEPHROTOXINS	• Aristolochic acid ○ Chinese herb nephropathy ○ Balkan nephropathy • Aminoglycosides (Amikacin, Tobramycin, Gentamycin) ○ Causes Non-oliguric AKI ○ Damages Collecting duct of Kidneys (V_2 receptors are affected impairing absorption of water) • Amphotericin B • Cisplatin • Contrast-Induced Nephropathy (CIN) ○ Contrast has direct toxic effects on renal tubules ○ Non-oliguric AKI with ↑ S. Creatine 1-2 days after the procedure, peaked by 3-5 days and resolve by 7 days ○ Metformin should be stopped 48 hours before contrast study in diabetics with $eGFR < 60$ ml/min d/t risk of developing lactic acidosis ○ Treatment → Hydration with IVF → Sodium bicarbonate (alkalinize renal tubule and prevent renal tubular injury) → Statins and N-Acetylcysteine: data inconclusive about beneficial effects → RRT
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- Rhabdomyolysis:
 - Labs: $\uparrow$ CPK MM and **tea/ red colored urine**
 - Causes:
 - Crush injury (release of myoglobin clogs the tubules)
 - Prolonged immobilization
 - Prolonged seizures: **GCSE** (prolonged seizures >5 mins)
 - Heat stroke
 - Malignant hyperthermia (Succinylcholine, Halothane): Core temperature $> 40.5^\circ\text{C}$
- Mismatched Blood Transfusion:
 - Type II hypersensitivity
 - Cause: Flank pain, Hemoglobinuria.
- Tumor Lysis Syndrome (TLS)- E.g. Burkitt's lymphoma



Mnemonic: **PUKE Calcium**

- Manifestations (Post-chemo):
 - **P** - $\uparrow \text{PO}_4$ due to ATP breakdown
 - **U** - $\uparrow$ Uric acid due to DNA breakdown by purines
 - **K** - $\uparrow \text{K}^+$
 - **Calcium** - $\downarrow \text{Ca}^{2+}$
- Prevention of TLS: Adequate pre-hydration with IVF
- Treatment: IVF + Furosemide (forced diuresis) + Allopurinol/Rasburicase
 - For TLS Rasburicase $>$ Allopurinol

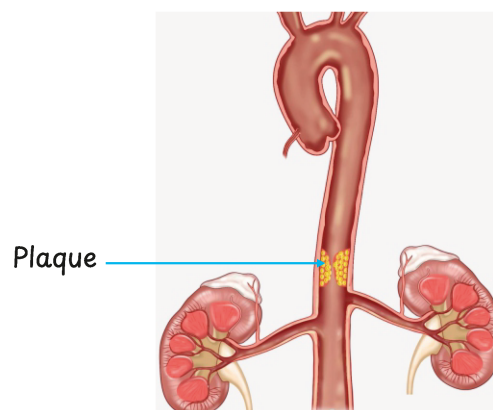
Important Information

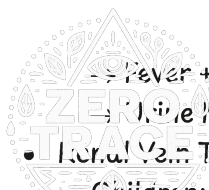
- **Ethylene glycol, and Diethylene glycol** (adulterant in cough syrup) $\rightarrow$ Formation of Calcium oxalate monohydrate crystals causing damage to tubules
- **Tumor lysis syndrome** $\rightarrow$ Uric acid crystals

VASCULAR CAUSES OF RENAL TYPE OF AKI

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- Atheroembolic Kidney Disease
 - Pre-procedure (Angiography) $\rightarrow$ **Normal** S. Creatinine
 - Post-angiography:
 - $\uparrow$ S. Creatinine
 - Livedo reticularis, Cholesterol embolism on fundus Ex
 - Urine M/E: **Eosinophiluria**
- Acute Interstitial Nephritis (AIN)
 - Causative Drugs:
 - Vancomycin
 - Methicillin
 - PPIs
 - NSAIDs
 - Features post exposure:



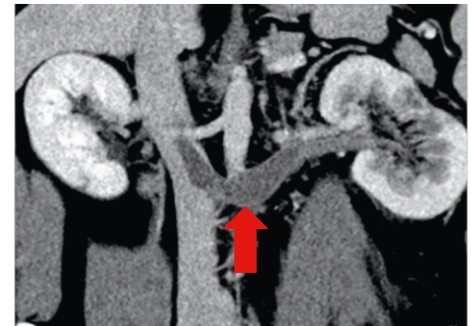


→ Fever + Rash + AKI

M/c M/E: **Eosinophiluria**

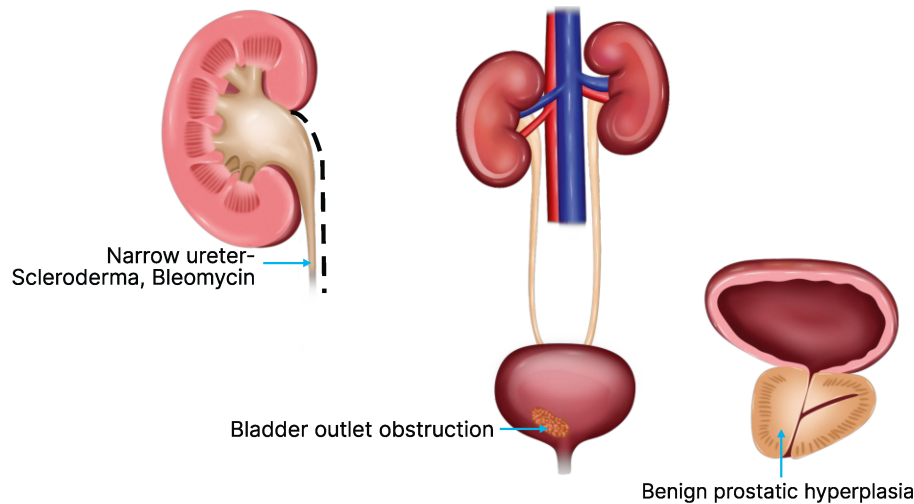
• Renal Vein Thrombosis

- Children: Diarrhea
- Adults: MGN (urinary loss of protein C, S and Antithrombin III)
- These patients develop
 - AKI, Hematuria
 - U/L kidney enlargement (painful)
- Investigations
 - Doppler: first investigation
 - IOC: **Renal venography**
- Complication: **Pulmonary embolism**



III. POST-RENAL AKI

- M/c cause: BPH
- Bladder outlet obstruction
- Scleroderma (Anti topoisomerase Ab), Bleomycin toxicity



CLINICAL FEATURES OF AKI

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- Nausea/vomiting, hiccups
 - ↑ Urea → irritation of
 - Diaphragm: Cause hiccups
 - Stomach: Cause vomiting
- HTN crisis: ↓ GFR → ↑ Renin
- Volume expansion causing anasarca / puffy eyes and pulmonary edema
 - D/t RAAS activation
- Oliguria/Anuria (50–100 mL/day)
- Chest pain (Uremic pericarditis → Hemoperitoneum)
- Decreased sensorium (d/t fluid shift across the brain), asterixis
- Bleeding (platelet dysfunction)
- Medication history (**ACE inhibitors / ARBs, Herbal medicines, Amikacin**)
- Past medical history



- Diarrhea
- Dysnatremia
- Contrast studies
- Angiography
- Poor urinary stream (BPH)
- Autoimmune disease (Scleroderma)

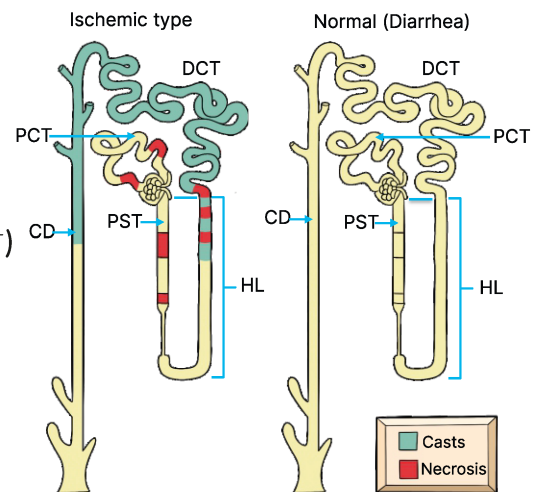
WORK UP

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- KFT: BUN↑, S. Creatinine ↑
- Serum electrolytes
 - ↓ Na⁺: Hypervolemic Hyponatremia
 - ↑ K⁺
 - ↓ Cl⁻ d/t vomiting
 - ↑ PO₄ and ↓ Ca²⁺

CAUSES OF HAGMA

- Ketoacidosis
- Uraemia
- Lactic acidosis
- Toxins
 - Uric Acid ↑↑ in Tumor Lysis Syndrome
- ABG - Metabolic acidosis/HAGMA
 - pH <7.2
 - HCO₃⁻ <15
 - Sodium bicarbonate correction: $0.5 \times \text{wt} \times (24 - \text{actual HCO}_3^-)$
- Urinary microscopic examination
 - Muddy brown granular casts: ATN
 - Dysmorphic RBC: Hematuria (AGN/PSGN/RPGN)
 - Eosinophiluria: Atheroembolic Kidney disease, Acute interstitial nephritis
- Urine osmolarity
 - Diarrhea → Loss of water and salt → Compensation by reabsorption → Urine Na⁺ is low
 - ATN → Damage to lining of tubules → Loss of absorption of salt, sugar, water → Urine Na⁺ is high



PRE-RENAL AKI	ISCHEMIC AKI
• Urinary sodium: Low	• Urinary sodium: High
• FeNa <1%	• FeNa >1%
• Urine osmolarity: >500 mOsm	• Urine osmolarity: 300 mOsm

- FeNa (<1%) can be falsely low in
 - Acute Glomerulonephritis
 - Contrast induced nephropathy
 - Rhabdomyolysis

- Ultrasound: To rule out Post-renal causes
- Chest X-ray
- Blood Culture
- ↓ ADAMTS13: TTP
- Autoimmune panel: Scleroderma
- CPK-MM: Rhabdomyolysis



Urine in Rhabdomyolysis

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TREATMENT

- Fluid management
 - Input-Output charting
 - Insensible losses = 800 ml/day
 - Trans epidermal loss (skin)
 - Water vapor (breathing)
 - Input fluids = Urinary output + Insensible loss
 - Fluids for maintenance in AKI
 - ↓Cl⁻ → Normal saline
 - Normal Cl⁻ → Ringer lactate
 - Sodium bicarbonate + Normal saline → **Forced alkaline diuresis**, to prevent tubular injury in **Rhabdomyolysis**
- Furosemide- Bolus, Drip
- Metabolic acidosis correction
- Hyponatremia: Fluid restriction + Diuretics
- ↑K⁺:
 - Restrict dietary potassium intake
 - Stop ACEI or ARBs
 - Calcium gluconate, Insulin drip (regular), Salbutamol nebulization
- Po₄↑
 - Dietary restriction
 - Calcium acetate
 - Sevelamer
 - Al(OH)₃ with meals
- 7. Hypertension crisis
 - IV Labetalol
 - Scleroderma crisis: ACE inhibitors
- 8. Bleeding risk
 - Desmopressin
 - Hemodialysis

INDICATIONS OF HEMODIALYSIS

Mnemonic : **AEIOU**

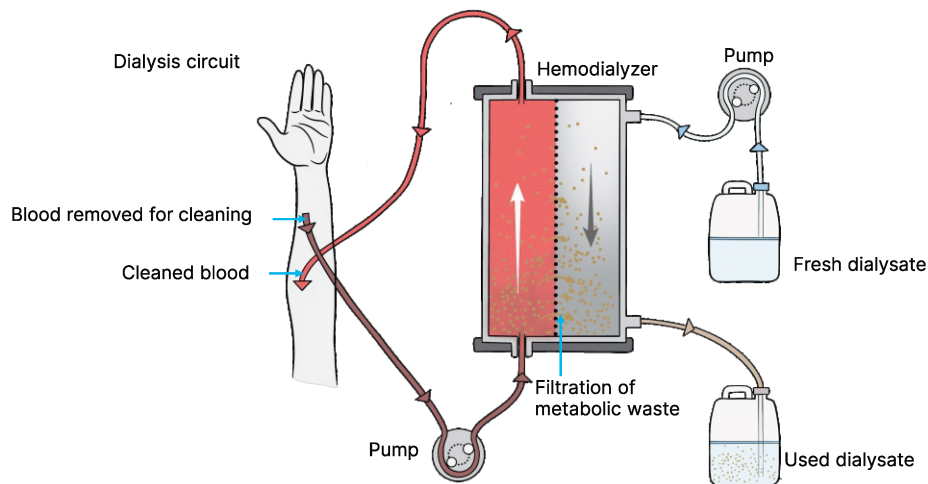
- **A**cidosis: pH <7.1 in spite of correction
- **E**lectrolyte: ↑ K⁺ in spite correction
- **I**nsufficiency: eGFR <10 ml/min



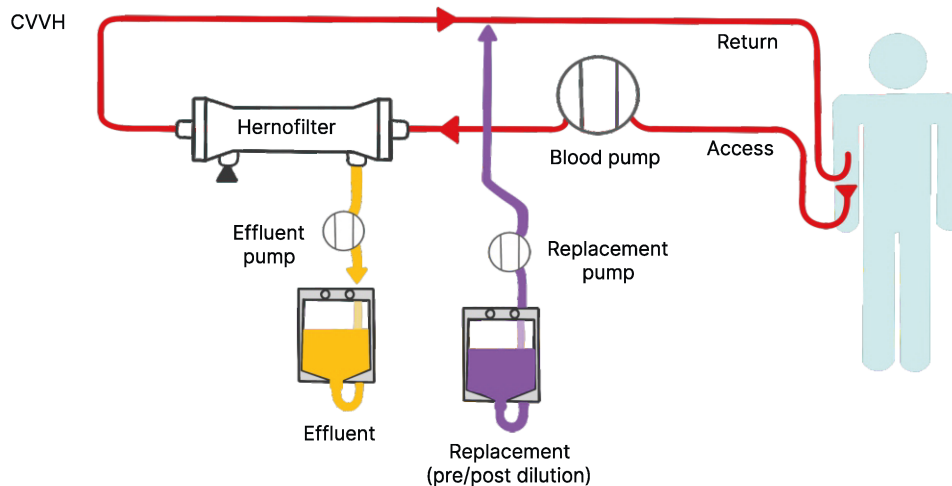
- Overload ↑ in spite of medication
- 4th pericarditis (absolute indication), Encephalopathy

INTERMITTENT HEMODIALYSIS

- Double lumen catheter: **Right internal jugular vein**, Subclavian vein, Femoral vein
- M/c complication: **Hypotension**
- Principle of HD:
 - Diffusion (solute clearance by concentration gradient) and ultrafiltration which is removal of water
 - Urea, Creatinine, and Potassium are removed



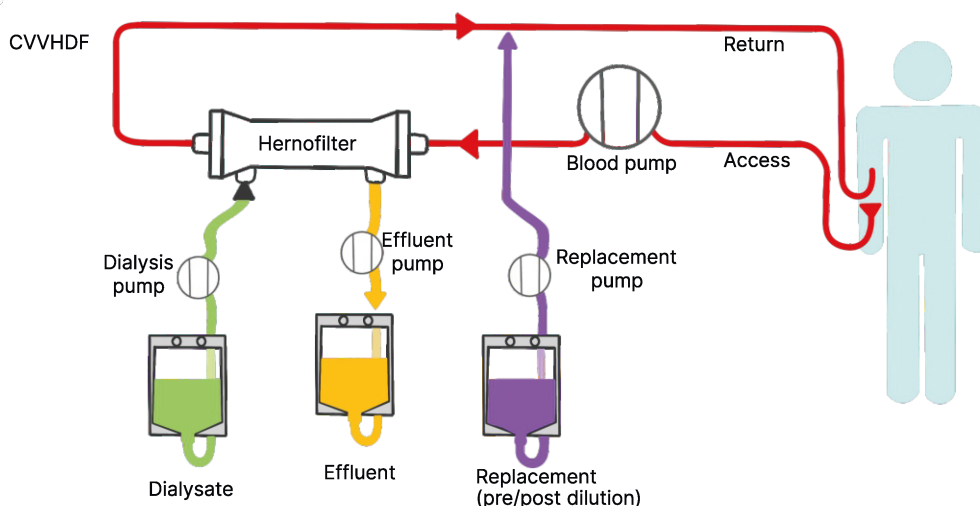
CVVH (CONTINUOUS VENOVENOUS HEMOFILTRATION)



- Principle: Convection (solvent drag) and is a slow continuous process over 24 hours where replacement fluid is added



CVVHDF (CONTINUOUS VENOVENOUS HEMODIAFILTRATION)



- Principle: Diffusion + convection. Max clearance of middle (myoglobin and cytokines) and small size molecules (U, C, K)
- Used if patient has severe hemodynamic instability/ cerebral edema

FEATURE	HEMODIALYSIS (HD)	CVVH (HEMOFILTRATION)	CVVHDF (HEMODIAFILTRATION)
Mode	Intermittent (3-4 hrs, 3x/week)	Continuous (~24 hrs/day)	Continuous (~24 hrs/day)
Solute Removal Mechanism	Diffusion (small molecules like urea)	Convection (removes larger molecules)	Both Diffusion + Convection
Fluid Removal	Limited ultrafiltration	More efficient volume removal	Most effective for volume & solute clearance
Efficiency	Rapid clearance, risk of hypotension	Slower but better tolerated	Best balance of solute & fluid removal
Indications	Stable CKD/ESRD patients, acute hyperkalemia, severe acidosis	Critically ill AKI patients with hemodynamic instability	Severe AKI with both fluid overload & solute clearance needs
Hemodynamic Stability	Risk of hypotension	Better tolerated in unstable patients	Best for unstable ICU patients
Protein Loss	Minimal	Higher (removes middle & large molecules)	Higher (but balances diffusion & convection)



Q. Which of the following tests is best to differentiate between AKI and CKD?

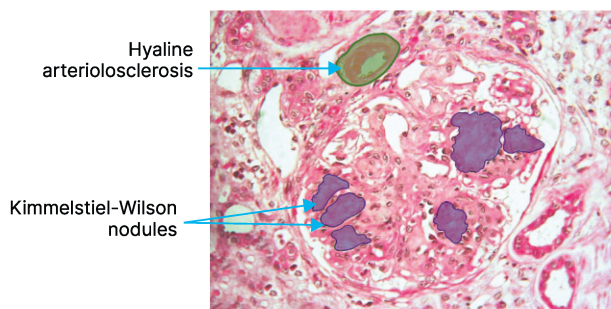
- a. Serum creatinine
- b. Size of kidney
- c. Albuminuria
- d. BUN

Ans. (c)

INTRODUCTION

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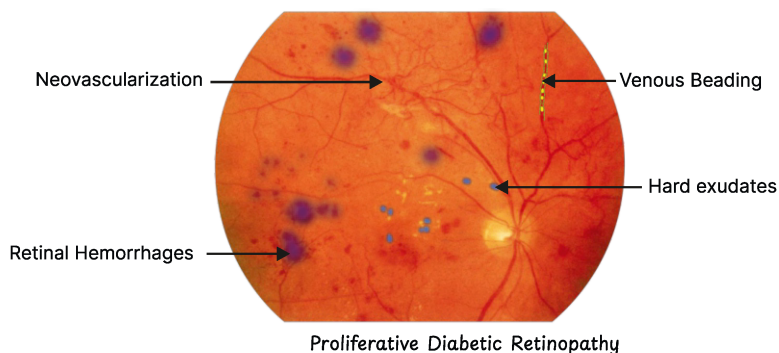
- Diabetic nephropathy
 - Classical HPE finding: **Nodular glomerular sclerosis (Kimmelstiel Wilson changes)**
 - Diabetic nephropathy and retinopathy often co-exist
 - Time to onset for both Diabetic nephropathy and retinopathy:
 - Type 1 DM: 5 years
 - Type 2 DM: 15-20 years
- Diabetic retinopathy



DIABETIC RETINOPATHY

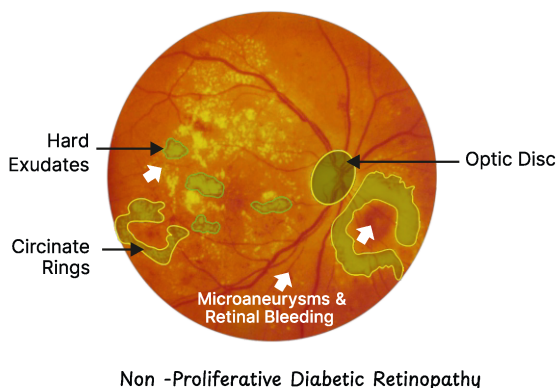
Proliferative diabetic retinopathy

- Features
 1. Neovascularization: earliest
 2. Retinal hemorrhages
 3. Venous bleed
 4. Hard exudates

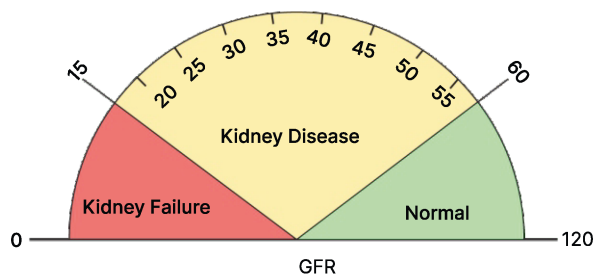


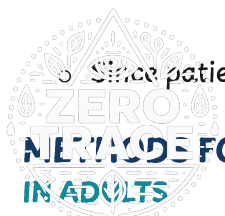
Non-Proliferative diabetic retinopathy

- Features
 1. Hard exudates
 2. Macular edema
 3. Circinate rings



- Definition: $eGFR < 60 \text{ mL/min for } \geq 3 \text{ months}$
- M/c cause: **Diabetic nephropathy**
- GFR varies with age and gender
 - It peaks at **120-125 mL/min around 30 years** of age and declines by 1 mL/min annually thereafter
- Serum creatinine
 - Varies with age, gender, and muscle mass





Since patients with CKD have muscle wasting, serum creatinine can be normal

NEW METHOD FOR GFR CALCULATION

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- CKD-EPI- Cystatin C equation
 - **Best method** for calculation
- Modification of Diet in Renal Disease (MDRD) Formula
- Cockcroft-Gault Equation $= \frac{(140 - \text{age}) \times \text{weight}}{72 \times \text{Serum creatinine}} = X$
 - For females, multiply 'X' by 0.85
 - Modified version of this formula is MDRD
- Creatinine clearance

IN CHILDREN

- Schwartz formula for GFR calculation in children
 - Height is considered instead of weight
 - Equation: $\frac{K \times \text{Height}}{\text{Serum Creatinine}}$
→ K = 0.41

Q. Calculate GFR of 5yr old child with weight = 16kg, height = 100cm and serum creatinine = 1mg/dl?

- a. 40ml/min
- b. 60ml/min
- c. 80ml/min
- d. 110ml/min

Solution:

$$\text{GFR} = \frac{0.41 \times 100}{1} = 41$$

Ans (a)

GFR GRADES

Grades	eGFR value in ml/min / 1.73 m ² BSA	Remarks
G1	≥90	Normal and high
G2	60-89	Mild
G3A	45-59	Mild-Moderate
G3B	30-44	Moderate - Severe
G4	15-29: Renal replacement therapy	Severe
G5	<15	Uremia/ ESRD/ Renal failure

URINARY ALBUMIN CREATININE RATIO (UACR)

As per sample, check for urine albumin creatinine ratio

A1	A2	A3
<30 mg/g	30-300 mg/g	≥300 mg/g
<3 mg/mmol	3-30 mg/mmol	≥30 mg/mmol

ALBUMIN AND GFR

Prognosis of CKD by GFR and albuminuria categories: KDIGO 2012				Persistent albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30mg/g >3 mg/mmol	30-300 mg/g 3-30 mg/mmol	>300 mg/g >30 mg/mmol
GFR categories (ml/min per 1.73m ²) Description and range	G1	Normal or high	≥90			
	G2	Mildly decreased	60-89			
	G3a	Mildly to Moderately decreased	45-59			
	G3b	Moderately to severely decreased	30-44			
	G4	Severely decreased	15-29			
	G5	Kidney Failure	<15			
Low risk (if no other markers of kidney disease, no CKD)				Moderately increased risk		High risk
				Very high risk		

CAUSES OF CKD

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Mnemonic: DG-HAT (accounts for 90% of cases)

- **D** - Diabetic Nephropathy
- **G** - Glomerulonephritis (CGN)
- **H** - Hypertension and Ischemic Heart Disease
 - In India IHD can cause ischemic cardiomyopathy → Drop in cardiac output → ↓ Blood supply to kidney → Progress into Cardiorenal syndrome
- **A** - Autosomal Dominant Polycystic Kidney Disease (ADPKD)
- **T** - Tubulointerstitial Kidney Diseases

Important Information

Bilateral Enlarged Kidneys

- DM
- ADPKD
- HIVAN (HIV associated Nephropathy)
- Amyloidosis

CLINICAL FEATURES OF CKD

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- Reduced appetite, weight loss
- Nausea, Hiccups (Urea accumulation → Irritates diaphragm)
- Volume expansion: Pedal edema and pulmonary edema
- Muscle cramps, Twitching, Irritability
 - ↓ Vitamin D₃ → ↓ Calcium
- Pruritis, Skin pigmentation
- Hypertension
- Evaluate for auditory, visual problems (Alport syndrome- COL4A gene defect- SNHL + Ant. Lenticonus + Renal failure) and Integumentary skin disease (Fabry's disease)
- Exposure to heavy metals

MANAGEMENT

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Mnemonic: ABCDE-K

HBA1c <7%

- Stop metformin if eGFR <30 mL/min (risk of lactic acidosis)
- Drugs that slow progression of Diabetic nephropathy
 - Low dose ACE inhibitor (1st line approach)
 - SGLT2 inhibitors: Empagliflozin
 - Aldosterone antagonist: Finerenone

Important Information

Anti-Diabetic drugs metabolized by Liver

- Linagliptin
 - Glipizide
-
- Insulin: 80% calculated dose should be given

BP CONTROL

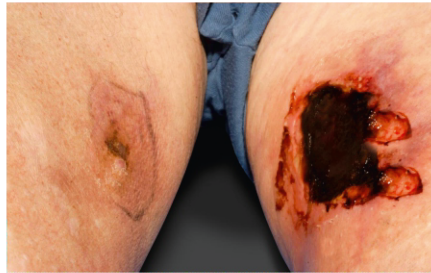
- Target BP <130/80 mmHg, if achievable with use of ACE inhibitors or ARBs
- Caution:
 - On starting ACE inhibitors, slight reduction in GFR, up to 30% reduction from the baseline may be seen
 - If progressive fall → Discontinue it
- Hypertension with Hyperkalemia in CKD
 - Lower K⁺ with Furosemide and Patiromer (novel K⁺ binder)
 - Stop ACEi, introduce CCB
 - Diuretics (Thiazide) safe in eGFR <30ml/min: Metazolone > Chlorthalidone (eGFR >30ml/min)
 - ESRD (eGFR <15ml/min): CCB + Alpha 1 blocker (Prazosin)

ELEVATED CHOLESTEROL

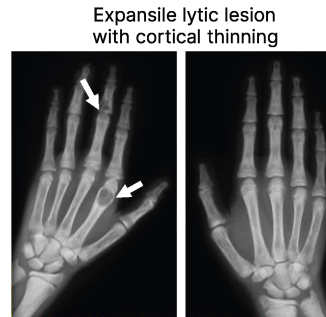
- Target LDL <70mg%
- Target LDL in a diabetic with previous h/o ASCVD (Post-Mi, CABG, PCI + Stenting) is <55mg/dl
- Highest-intensity Atorvastatin: 40-80mg/day

VITAMIN D IS LOW

- **Calcium** and causes 2° Hyperparathyroidism
- Vitamin D is synthesized in PCT of kidney
- PCT contains 1- α hydroxylase which helps in synthesis of 1,25-dihydroxycholecalciferol
- **$\uparrow\text{PO}_4$** : D/t inability to excrete phosphate
 - Tumoral calcinosis (lobular calcifications in peri-articular soft tissue)
 - Calciphylaxis: Painful livedo reticularis
 - Calcification of Iliac vessels
- **High bone turnover**- $\downarrow$ Calcium, $\uparrow\text{PO}_4$, $\uparrow$ Serum alkaline phosphatase
- **$\uparrow\text{PTH}$** : Osteitis cystica fibrosa / Brown tumor (Misnomer)
 - Hand X-ray: Subperiosteal resorption of phalanges



Calciphylaxis



Osteitis fibrosa cystica

MANAGEMENT OF HYPERPHOSPHATEMIA IN CKD

- PO_4 binders
 - **Sevelamer**, Lanthanum
 - Calcium acetate
- $\uparrow\text{PTH}$: Use Calcimimetic drugs
 - For PTH normalization
 - Drug: **Cinacalcet**

Important Information

SUMMARY OF ELECTROLYTE IMBALANCES SEEN IN CKD

- Sodium $\downarrow$
 - In this condition, there is $\uparrow$ Aldosterone $\rightarrow$ Hypervolemic Hyponatremia
- Potassium: Both Hyper and Hypokalemia can occur
 - **Hyperkalemia**: D/t inability to excrete potassium
 - **Hypokalemia**: D/t excess use of diuretics to treat edema
- PO_4 $\uparrow$
- Calcium $\downarrow$

ADYNAMIC BONE DISEASE IN CKD PATIENTS

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- A low bone turnover state due to suppressed osteoblast or osteoclast activity
- Causes:
 - Over suppression of PTH from excessive calcium acetate or calcium-rich dialysate use
 - Uremia-induced low bone response
- **Key feature**: Silent bone, no remodelling activity, leading to brittleness ("brittle as chalk")

ERYTHROPOIETIN DEFICIENCY

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- $\uparrow$ Reticulocyte count $\rightarrow$ Normocytic normochromic anemia
- Main cause anemia in CKD - Erythropoietin deficiency
- Elevated PTH: Causes bone marrow fibrosis
- Anemia of chronic disease
- Increased risk of bleeding

TREATMENT:

- Target Hb: 10-10.5 gms
- Initial T/t $\rightarrow$ IV iron
- ESA (Erythropoietin Stimulating Agents): S/E HTN
 - Darbepoetin
 - Epoetin
- HIF-PH stabilizer (Hypoxia-inducible factor Prolyl Hydroxylase inhibitor)
 - Roxadustat (oral)

HYPERKALEMIA

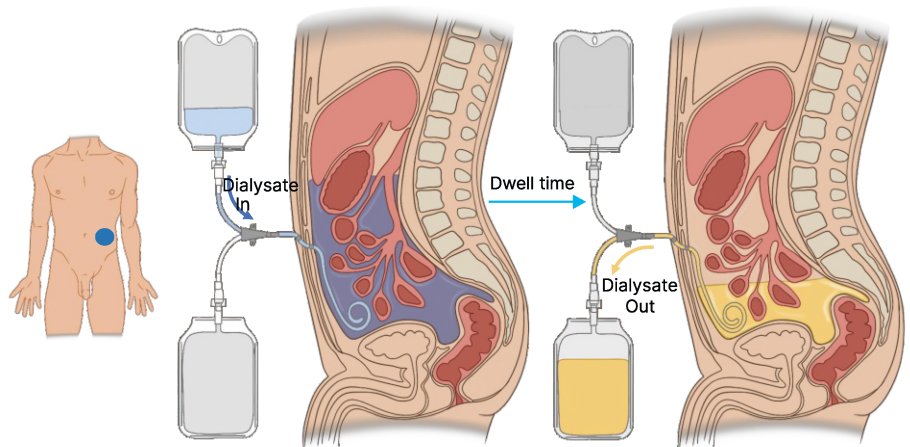
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- Diuretics recommended:
 - Initial choice: Thiazides
 - Furosemide + Metolazone $\rightarrow$ To reduce pedal edema
- Novel Potassium binders:
 - Patiromer
 - Sodium zirconate
 - SPS (Sodium Polystyrene Sulfonate)
- Hyperkalemia refractory to medical therapy $\rightarrow$ Hemodialysis

RENAL REPLACEMENT THERAPY

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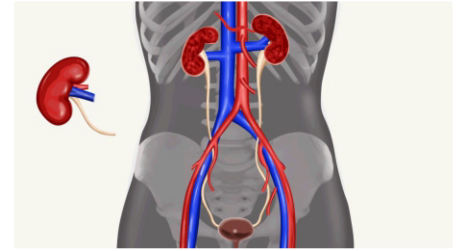
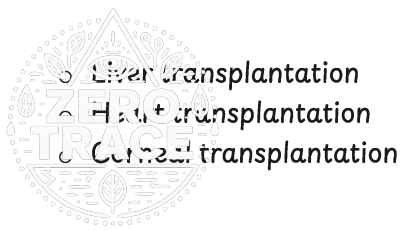
- Decision making by G4
- 1. Allogeneic Kidney Transplantation
 - HLA match, blood group match
- 2. Hemodialysis:
 - Each session: 4 hrs $\times$ 3 times/week
- 3. Peritoneal dialysis
 - Contraindicated in ADPKD
 - More suitable for pediatric age group as compared to adults



Peritoneal Dialysis

ALLOGENEIC KIDNEY TRANSPLANTATION

- Gene for HLA: Chromosome 6
- Gene for Blood Group: Chromosome 9
- HLA matching not mandatory for:



DONOR TESTING

- HIV
- HBV
- HCV
- Syphilis
- CMV (important: causes graft failure)
- Doppler: To rule out unilateral renal artery stenosis

RECIPIENT

- HLA matching for: **A, B, C, DP, DQ, DR**

DONOR		
HLA A	HLA B	DR
2	44	4
1	8	3

RECIPIENT		
HLA A	HLA B	DR
2	8	3
3	44	4

- Perform MRA before transplantation to rule out Iliac artery calcification
- If present, vascular anastomosis with new kidney cannot be made
- **Left kidney** of donor is used as Left renal vein is longer, offering better anastomosis
- It is placed in **Right iliac fossa** of recipient

Important Information

- Isograft or syngeneic transplantation occurs in monozygotic twins

MANAGEMENT

1/6 or 2/6 Match	3/6 Match	4/6 or 5/6 Match
Induction: Anti-thymocyte globulin (ATG)	Induction: Anti-CD2 molecule (Basiliximab)	No induction therapy needed

MAINTENANCE THERAPY (Life-long basis)

Mnemonic: **STM-A**

- **S**teroid
- **T**acrolimus / Cyclosporine
- **M**ycophenolate
- **A**zathioprine



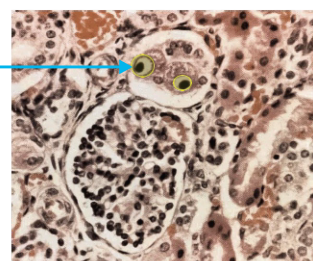
TYPES OF KIDNEY REJECTION

REJECTION	TIME	HYPERSENSITIVITY
Hyperacute	Minutes-Hours	Type II
Acute	<6 months	Type II and IV
Chronic	>6 months	Type IV

POST-TRANSPLANTATION OPPORTUNISTIC INFECTIONS

- <1 month
 - MRSA, Herpes, VZ
- 1-6 months
 - P. jiroveci, CMV: Graft failure
 - Urine M/E in CMV: Intranuclear/intracytoplasmic basophilic inclusions; "Owl eye" appearance
 - Hepatitis B/C
- >6 months
 - Aspergillus, Nocardia, BK virus (Polyomavirus, Decoy cell is seen)

Basophilic intranuclear inclusion body



Owl's Eye Appearance - CMV infection in Kidney

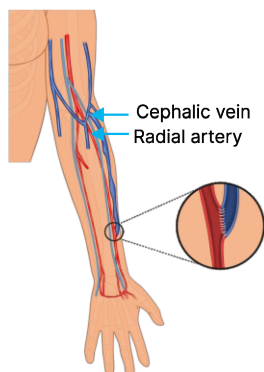
Extra Mile:

- M/c cause of graft failure after SOT or HSCT: CMV
 - Solid organ transplantation (SOT)
 - Hematopoietic stem cell transplantation (HSCT)
- M/c infection in Post-transplant lymphoma: **EBV**
- Cancer seen with maintenance therapy given after Kidney transplantation: **Skin cancer**

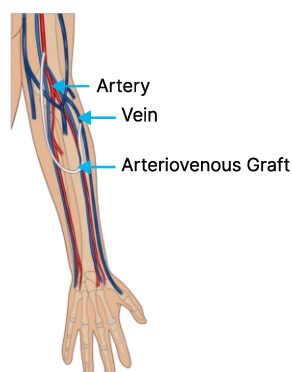
HEMODIALYSIS

01:00:52

- Extracorporeal: Membrane outside the body
 - Hollow fibre dialyser: Biocompatible
 - Earlier: Cellulose-based dialyzer
 - Activated complement system
 - ↑risk of anaphylactoid reactions
- Vascular Access:
 - **Brescia-Cimino Fistula**: Created between Radial artery and Cephalic vein
 - Arteriovenous Graft: PTFE



Radiocephalic fistula



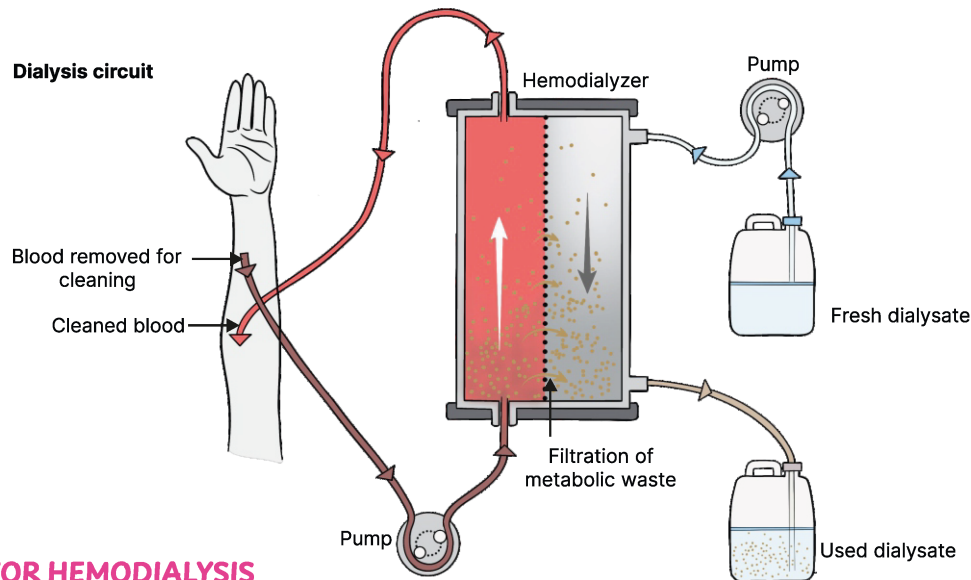
Forearm loop graft

RATIONALE FOR CREATION OF AV FISTULA

- Arterialization of vein to reduce the incidence of thrombophlebitis. Vein becomes thicker and stronger to tolerate repeated punctures and handle flow rates up to 600 ml/min
- Pressure gradient for free blood flow into the dialysis catheter
- Harvesting of fistula
 - Identified by bruit and doppler scan and takes 4–6 weeks

PRINCIPLES OF HEMODIALYSIS

1. Solute diffusion: Removes waste products (Urea, Creatinine, Potassium)
2. Solvent drag (ultrafiltration): Removes excess fluid



INDICATIONS FOR HEMODIALYSIS

Mnemonic: AEIOU

- **A**cidosis (Metabolic): pH <7.1, refractory to medical therapy
- **E**lectrolyte imbalance: Hyperkalemia refractory to medical therapy
- Renal **I**nsufficiency
 - eGFR <10 mL/min
 - eGFR <6 mL/min
- **O**verload volume: Pulmonary edema refractory to drugs
- **U**remic pericarditis/encephalopathy
 - Absolute indication

COMPLICATIONS

- M/c Acute complication: Hypotension
- M/c complication of recurrent HD: Accelerated atherosclerosis (CV mortality ↑)
- Other complications
 - Dialysis dementia: $A\beta_2$ Amyloid deposition
 - Carpal tunnel syndrome: $A\beta_2$ Amyloid deposition
 - Dialysis disequilibrium
 - Seen when dialysis done first time in CKD
 - Manage with mannitol
 - Peripheral neuropathy



Q. GFR in ml / min for a client with chronic kidney disease grade 3A is?

- a. 60-89
- b. 45-59
- c. 30-44
- d. 15-29

Ans. (b)

Q. A 35-year-old male patient who underwent kidney transplantation last year presents with features of graft failure. Urine microscopy image is shown below. Diagnosis is?

- a. BK virus
- b. Pneumocystis jirovecii
- c. Cytomegalovirus
- d. Hepatitis B

Ans. (a)

